Assignement 3
LSE, PCA and PCR model identification method

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Contents

1. Introduction 3
2. Theoretical Background 4
   2.1. Principal Component Analysis (PCA) 4
   2.2. Principal Component Regression (PCR) 4
3. Implementation 5
   3.1. PCA 5
   3.2. PCR 5
4. The impact of data standardization on results 7
   4.1. LSE 7
   4.2. PCA 8
   4.3. LSE vs PCA 10
5. The problem of collinearity 12
   5.1. Implementation of the loop 12
      5.1.1. Collinearity of the parameters 12
   5.2. Results 12
      5.2.1. LSE 13
      5.2.2. PCR 15
Bibliography 17
Index of Figures 18

1. Introduction

This report explores the application of Least Squares Estimation (LSE), Principal Component Analysis (PCA), and Principal Component Regression (PCR) for model identification. The goal is to identify an equation to calculate the inverse of the yield based on measured variables: cooling air flow (A_FLOW), water temperature (T_H2O), and acid concentration (C_ACID).

2. Theoretical Background

2.1. Principal Component Analysis (PCA)

PCA is a technique used to emphasize variation and bring out strong patterns in a dataset.

Principal Component Analysis (PCA) is a widely used statistical technique for dimensionality reduction while preserving as much variability as possible in the data. It transforms the data into a new coordinate system where the greatest variance lies on the first coordinate (principal component), the second greatest variance on the second coordinate, and so on. Here's a step-by-step explanation of how PCA has been implemented in this assignement:

0. Standardization: The topic of the first part of the assignement is the impact of standardization (and the absence thereof) on PCA. As such, it is not included inside the implementation of the algorithm.

1. Covariance Matrix Computation: Computing the covariance matrix of the data to captures the variance between different variables and helps in identifying how the variables relate to each other.

2. Covariance decomposition: Performing decomposition on the covariance matrix to obtain the eigenvalues and eigenvectors inherent of the data set. The eigenvalues represent the amount of variance carried in each principal component, while the eigenvectors (principal components) determine the directions of the new feature space.

3. Sorting Eigenvalues: Sorting the eigenvalues in descending order to rank the corresponding eigenvectors. This helps in selecting the top eigenvectors that represent the most variance in the data.

4. Parameter determination: The matching parameters are determined by the following formula similar to LSE :

$$\theta = P(T^T T)^{-1} T^T y \tag{1}$$

where T is the scores vector, P the principal component matrix and y the system output.

2.2. Principal Component Regression (PCR)

PCR involves using PCA and LSE in combination to reduce the dimensionality of the data before applying linear regression. Two additional steps are introduced between 3. and 4. steps :

1. Selecting the principal components: Once PCA is applied, eigenvalues and eigenvectors are computed into cumulative sums to determine if their ratios exceed a threshold, indicating their importance relative to the dataset.

2. Creating and projecting the dataset inside principal components space: PCA results and initial dataset is transformed into the reduced space before parameter estimation.

After parameter estimation, they are projected back into the original space.

3. Implementation

3.1. PCA

PCA :

```
function [params, ndata] = PCA(data, y)
    [kcoeff, kscore, klatent] = PCA_om(data);
    s_num = estimatePrincipal(klatent, 0.95); % Principal component selection.
    params = (data*kcoeff(:, 1:s_num))\y; % Data proj. & LSE
    ndata = data*kcoeff(:, 1:s_num);
end
```

Core PCA code :

```
function [pc, X_pca, latent] = PCA_om(data)
    [n, m] = size(data);
    data = data - mean(data);%, 'all');

    cov_mat = (data' * data)/(n-1); % Computing the cov matrix

    [eigVec, eigVal] = svd(cov_mat); % Computing eigenvectors and eigenvalues
    [eigVal_sort, ind] = sort(diag(eigVal), 'descend');
    eigVec_sort = eigVec(:, ind);

    X_pca = data * eigVec_sort(:, 1:m);
    pc = eigVec_sort(:, 1:m);
    latent = eigVal_sort(1:m);
end
```

3.2. PCR

PCR :

```
function params = PCR(data, y)
    [kcoeff, kscore, klatent] = PCA_om(data); % Initial parameters est.
    s_num = estimatePrincipal(klatent, 0.95); % Principal component selection.
    [Ps, Xs, Ts] = approxData(kcoeff, data, s_num); % Projection into principal
component space
    params = approximateParamsPCR(Xs, y, Ps, Ts); % Est. of parameters with cast
back into main space
end
```

estimatePrincipal :

```
function s_num = estimatePrincipal(eigVal, threshold)
    explained_variance = cumsum(eigVal) / sum(eigVal); % Cumulative variance
explained
    s_num = find(explained_variance >= threshold, 1); % Choose K PCs explaining
95% variance
end
```

ApproxData :

```
function [Ps, Xs, Ts] = approxData(P, X, s)
    Ps = P(:, 1:s);
    Ts = X * Ps;
    Xs = Ts * Ps';
end
```

PCR parameters estimation :

```
function approxParams = approximateParamsPCR(X, Y, P, T)
    % Project the data onto the principal component space
    T = X * P;

    % Estimate parameters in the principal component space
    beta_PC = (T' * T) \ (T' * Y);

    % Transform parameters back to the original feature space
    approxParams = P * beta_PC;
end
```

4. The impact of data standardization on results

4.1. LSE

LSE	No std	Std	Z-score
σ of error	3.7535	0.1072	0.3690
Avg of error	−2.7008	−0.0772	−0.2655
Theoretical var	0.0051 0.0379 0.0068 39.6142	0.0037 0.0031 0.0025 0.0010	0.0041 0.0037 0.0019 0.0014

This table of data has been obtained on the data used during training.

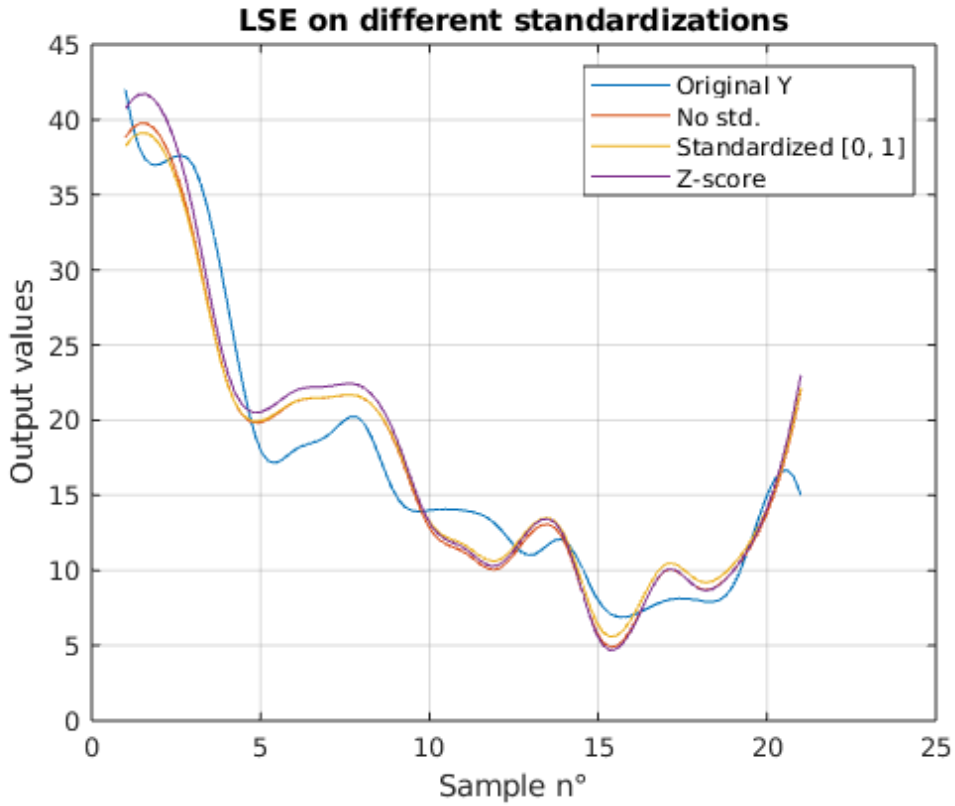


Figure 1: LSE on different datasets

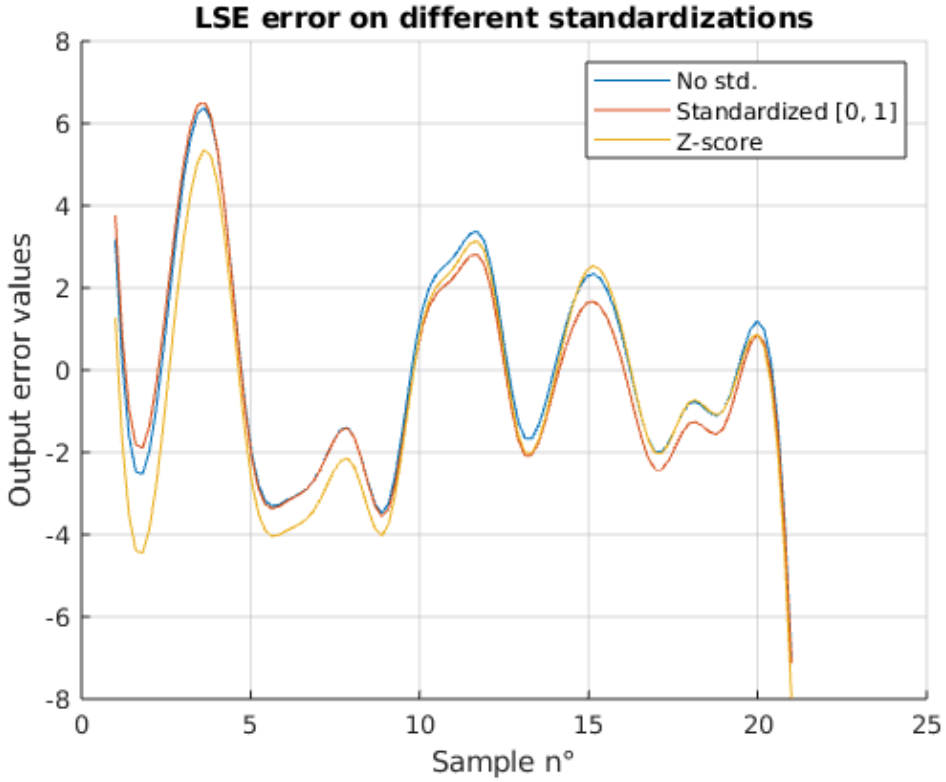


Figure 2: Output error of estimation after LSE

The original dataset has been interpolated with *spline* method, the idea being to test LSE output on new data that is not the training data while keeping the same shape.

The effect of different standardization appears visually in this graph: while keeping the same dynamics, LSE on non-standardized data is the worse of the three cases, [0, 1] standard and z-score being better and really similar.

The standard deviation and average of the error show that not standardizing data result in greater error.

4.2. PCA

PCA	No std	Std	Z-score
σ of error	1.3142	0.0650	0.2235
Avg of error	−32.2865	−0.0204	−0.0357

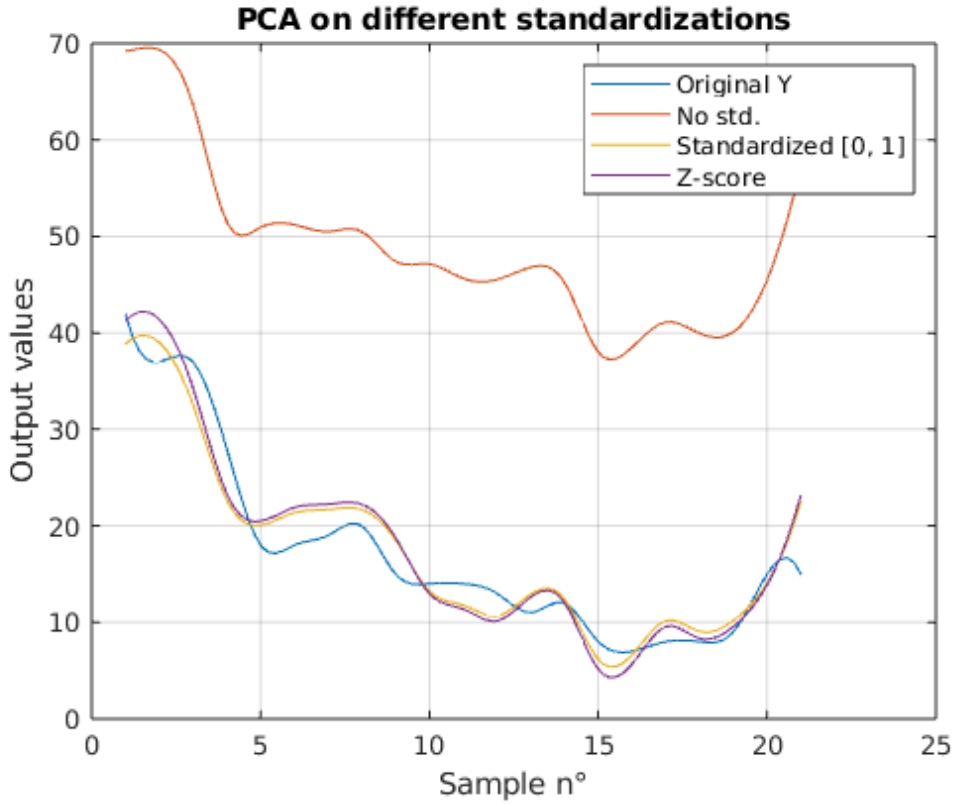


Figure 3: Parameter estimation after PCA

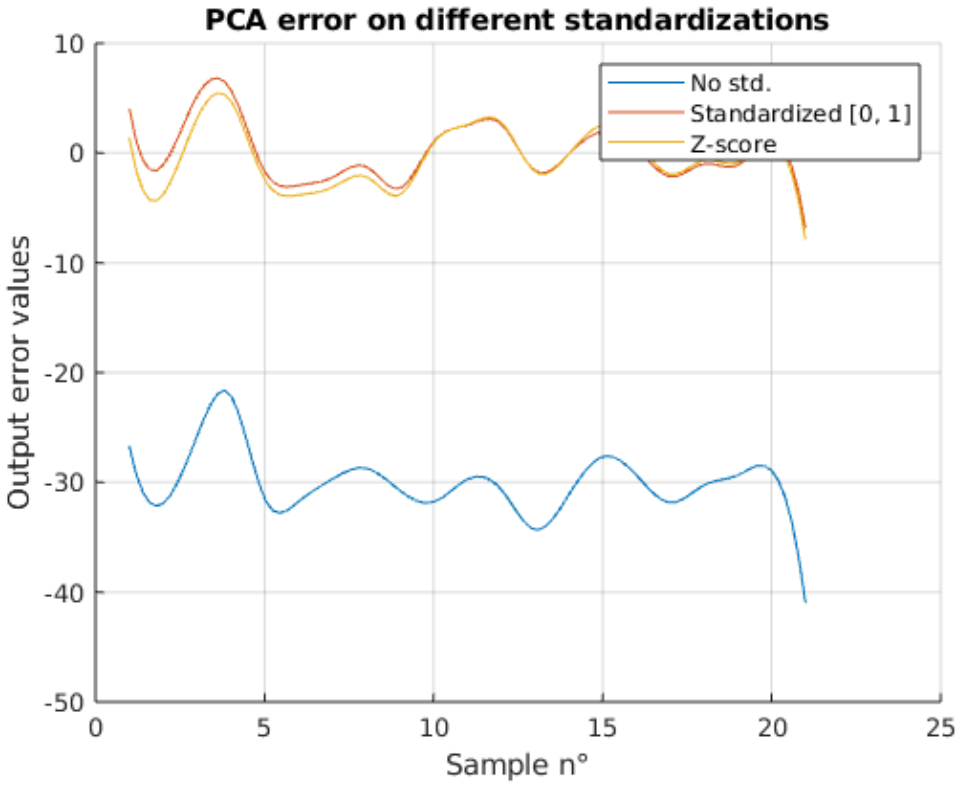


Figure 4: Output error of estimation after PCA

With PCA, the z-score and [0, 1] datasets perform slightly better than their LSE counterparts (while being very similar in shape), showing a slightly better conditioning.

However, this is not the case with unstandardized data, with it being really off the charts. It presents the same dynamics as the other models, but with an enormous bias of around −30. This is due to the covariance matrix being sensitive to big numbers, skewing the eigenvalues and the result.

4.3. LSE vs PCA

	LSE	PCA
σ of error	2.5431	2.5200
Avg of error	−0.1064	−0.3145

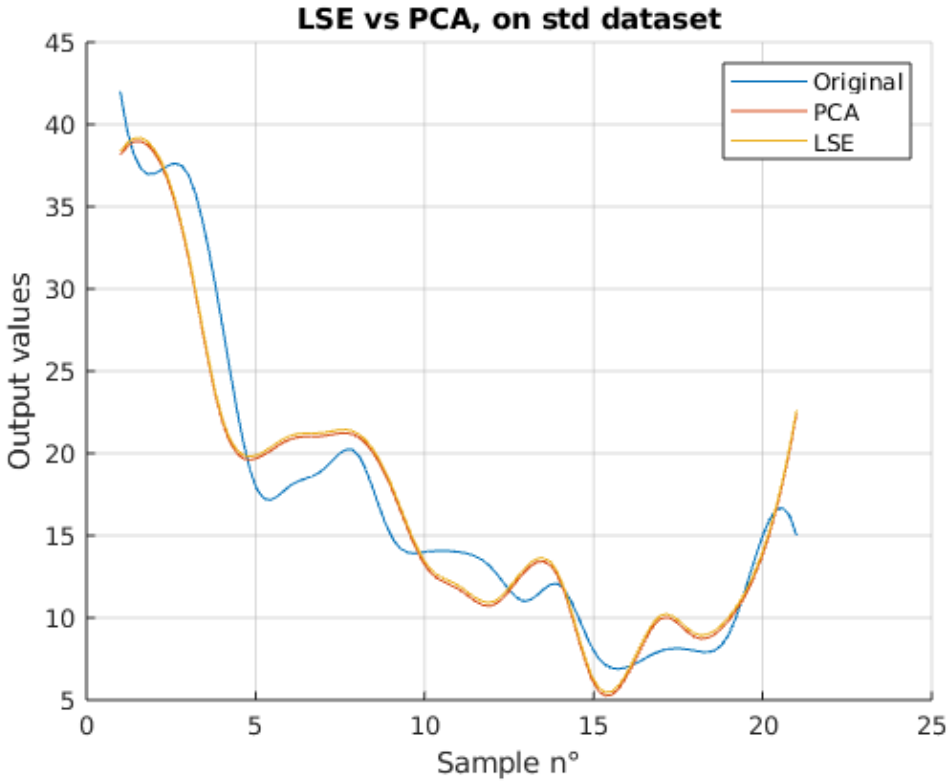


Figure 5: Parameter estimation after PCA

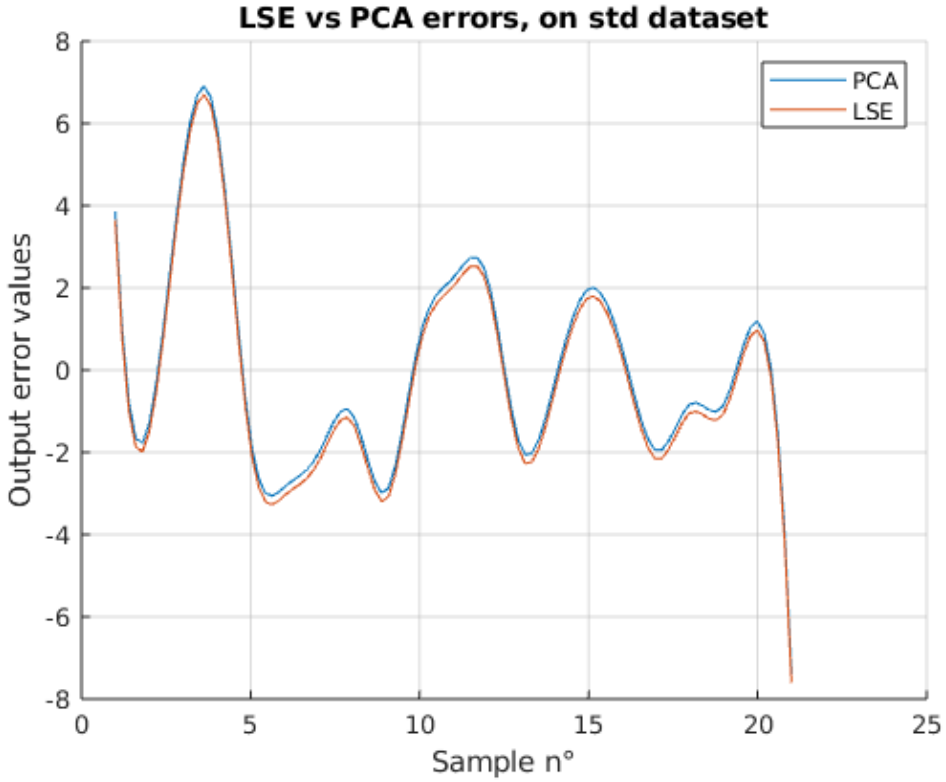


Figure 6: Output error of estimation after PCA

These 2 methods are really similar to one another when compared, with the tradeoff of better standard deviation of the error for PCA for a worse average error.

5. The problem of collinearity

5.1. Implementation of the loop

```
X2 = [X, 2*T_H2O+6+0.1*randn(size(T_H2O))];
Y2_lse = Y;
Y2_pcr = Y;

params_lse_cumul = []; %[0; 0; 0; 0];
params_pcr_cumul = [];%[0; 0; 0; 0];

for k=1:10
    X2 = [X, 2*T_H2O+6+0.1*randn(size(T_H2O))];
    params_lse = nLSE(X2, Y2_lse, alpha);
    params_pcr = PCR(X2, Y2_pcr);

    params_lse_cumul = [params_lse_cumul, params_lse];
    params_pcr_cumul = [params_pcr_cumul, params_pcr];

    Y2_lse = X2*params_lse;
    Y2_pcr = X2*params_pcr;
end
```

X2, which is the new collinear data set, is recalculated at each iteration. At every step, new Y used for parameter estimations are calculated from the new X2 and previous parameters. On top of that, parameters are calculated on these new Y2.

5.1.1. Collinearity of the parameters

To visualize the collinearity between parameters, we use the correlation matrix provided by MATLAB. For this particular case :

	A_FLOW	T_H20	C_ACID	X2
A_FLOW	1	0.7819	0.5001	0.7853
T_H20	0.7819	1	0.3909	0.9999
C_ACID	0.5001	0.3909	1	0.3908
X2	0.7853	0.9999	0.3908	1

As we can see, the newly imported X2 data is so strongly correlated with T_H2O that we can essentially consider X2 as T_H2O with added noise. We can expect LSE to stumble a bit on this strong correlation, while almost no changes for PCR that will select T_H20 against X2.

5.2. Results

	LSE at 0	LSE at 10	PCR at 0	PCR at 10
σ of error	2.9968	2.9902	3.9554	3.9520
Avg of error	0.0073	0	−0.1457	−0.1483
Var of the parameters	1.1104e+03	326.5169	0.2961	0.2947

Theoretical variance	1.0e+04 x (0.0000 0.3179 0.0000 7.0309 0.0888)	1.0e+04 x (0.0000 0.0197 0.0000 1.8974 0.0888)		

5.2.1. LSE

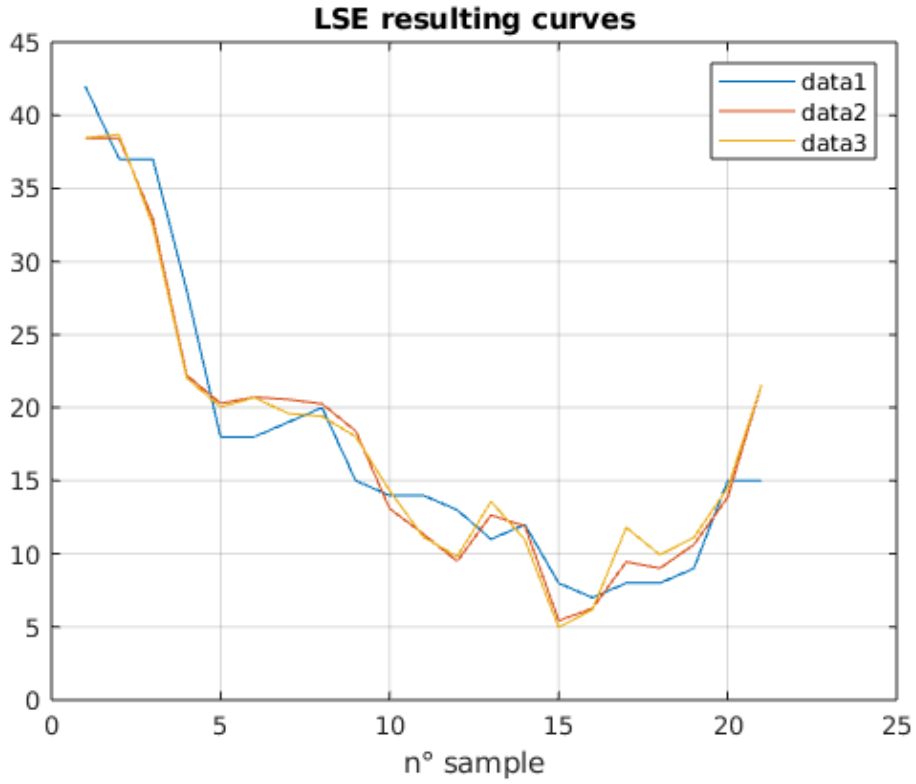


Figure 7: Comparison between Y, LSE in the beginning (data2) and after (data3)

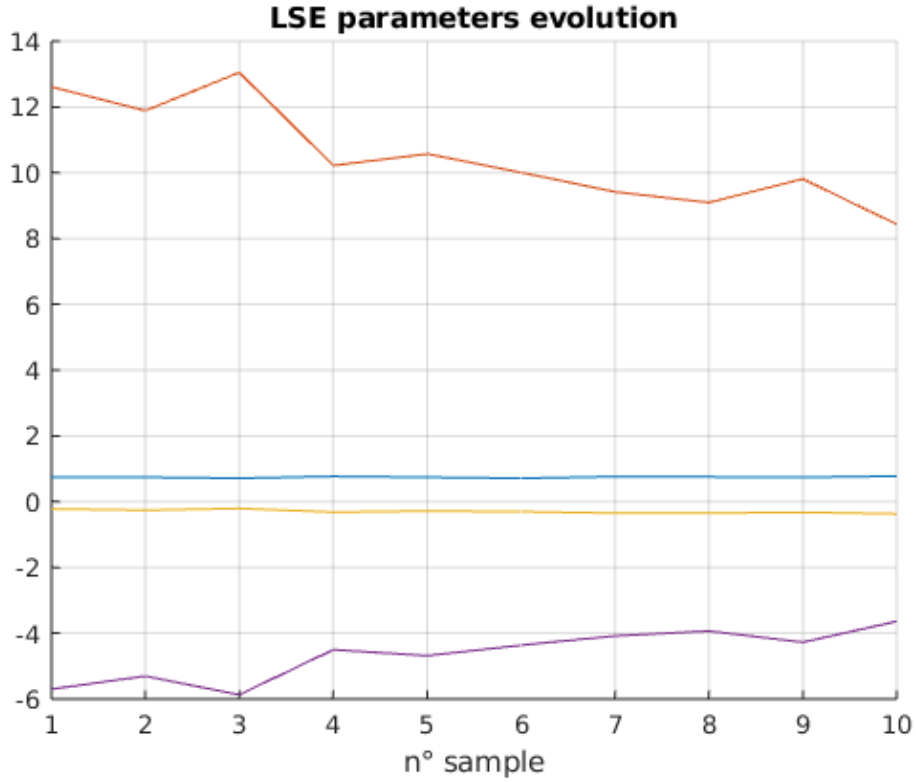


Figure 8: Parameter evolution after 10 iterations, T_H20 in orange X2 in violet

The parameter evolution graph highlights the convergence of the parameter towards a certain point. While the parameter depending on C_ACID and A_FLOW show almost no change (except for the noise), the two depending on T_H20 and X2 bear strong evolutions and a high variance. As a result, the parameter variances at every time of the loop are really high.

5.2.2. PCR

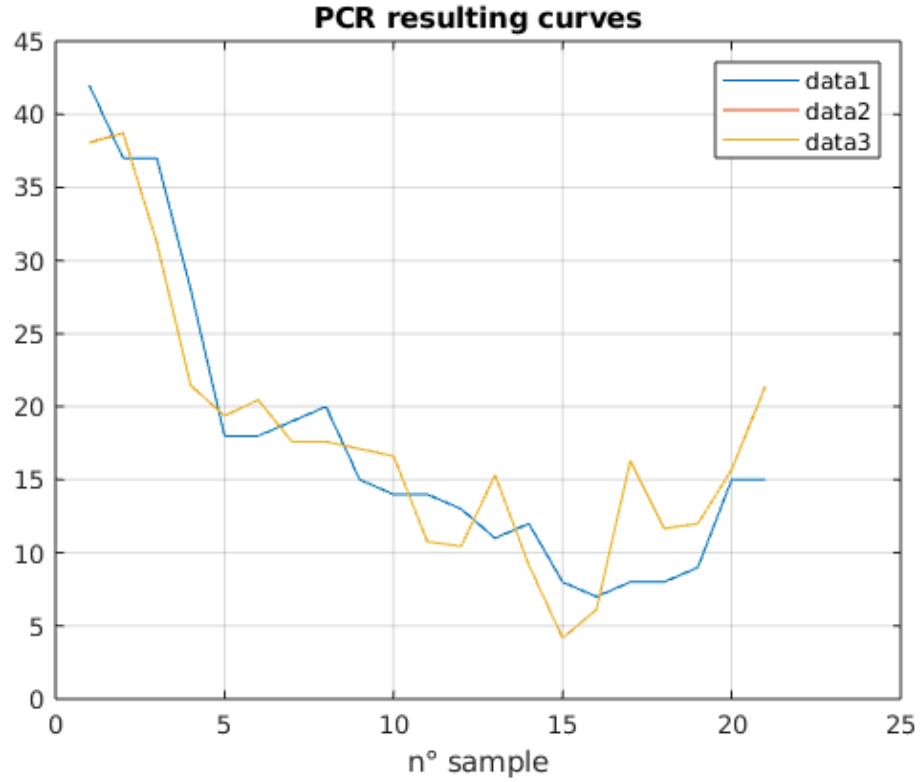


Figure 9: Comparison between Y, PCR in the beginning (data2) and after (data3)

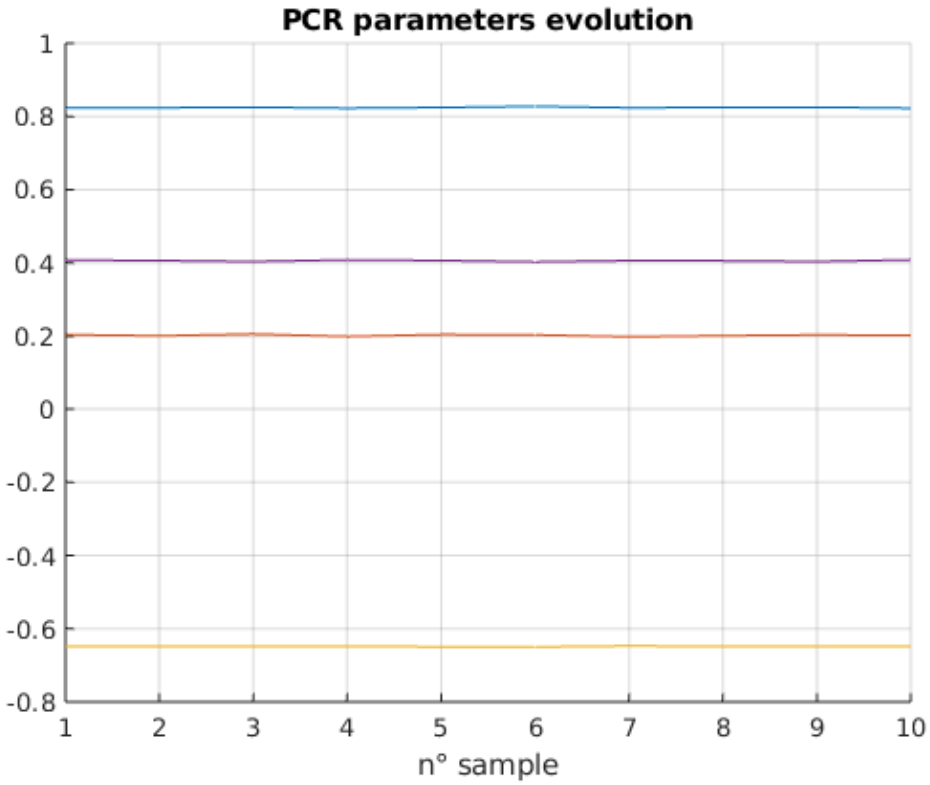


Figure 10: Parameter evolution after 10 iterations

As announced, T_H20 is selected over X2, resulting in the parameters only changing for the noise that we introduce. The parameter evolution graphs shows no strong evolution,

confirming this fact. The comparison between the first iteration and the second iteration also highlights this property from PCR.

The variance throughout the iterations is really small, from around 0.3830 to 0.3840, or $\sigma_p^2 = 0.01^2$ the actual amplitude of the noise. It indicates the resilience of the PCR algorithm over LSE towards the presence of strongly correlated variables.

The performance is a little lower however, with a MSE of around $e = 14.9641$, or an increase of 61.2%.

Bibliography

Index of Figures

Figure 1 LSE on different datasets 7
Figure 2 Output error of estimation after LSE 8
Figure 3 Parameter estimation after PCA 9
Figure 4 Output error of estimation after PCA 9
Figure 5 Parameter estimation after PCA 10
Figure 6 Output error of estimation after PCA 11
Figure 7 Comparison between Y, LSE in the beginning (data2) and after (data3) 13
Figure 8 Parameter evolution after 10 iterations, T_H20 in orange X2 in violet 14
Figure 9 Comparison between Y, PCR in the beginning (data2) and after (data3) 15
Figure 10 Parameter evolution after 10 iterations 15